

MANAGEMENT OF ACUTE EXACERBATION OF ASTHMA C-II

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Outline

- Introduction
- Epidemiology
- Etiologies and Pathogenesis
- Wheezing phenotype
- Types of childhood asthma
- Clinical manifestation
- Diagnosis
- Management

Objectives

- At the end of this presentation you are expected to
- Know about etiologies and pathogenesis of Asthma
- Classification , Clinical manifestation and diagnosis of asthma
- Management of acute exacerbation of asthma

Introduction

- Asthma is a heterogeneous disease, usually characterized by chronic airway inflammation.
- It is defined by the history of respiratory symptoms, such as wheeze, shortness of breath, chest tightness and cough, that vary over time and in intensity, together with variable expiratory airflow limitation

- Clinical asthma phenotypes

Allergic asthma

Non allergic asthma

Cough variant asthma and cough
predominant asthma

Adult onset asthma

Asthma with persistent air flow limitation

Asthma with obesity

EPIDEMIOLOGY

- Asthma is a common chronic disease, causing considerable morbidity, it affects 1-29% of world population.
- Increasing by 50% per decade worldwide, despite advanced treatments. Current wheeze affects 2.4%–37.6% of 6–7-year-olds and 0.8%–32.6% of 13–14-year-olds globally.
- **The Allergic Triad:** Asthma prevalence strongly correlates with rates of eczema and allergic rhinitis..

- Urban vs. Rural Divide: Highly prevalent in modern, affluent cities; significantly lower in rural areas and farming communities with animals.
- Approximately 80% of all asthmatic patients report disease onset prior to 6 yr. of age. However, of all young children who experience recurrent wheezing, only a minority go on to have persistent asthma in later childhood

Pathogenesis

- Airflow obstruction in asthma is the result of numerous pathologic processes. In the small airways, airflow is regulated by smooth muscle encircling the airways lumens; bronchoconstriction of these bronchiolar muscular bands restricts or blocks airflow.
- A cellular inflammatory infiltrate and exudates distinguished by eosinophils, but also including other inflammatory cell types (neutrophils, monocytes, lymphocytes, mast cells, basophils), can fill and obstruct the airways and induce epithelial damage and desquamation into the airways lumen.
- Helper T lymphocytes and other immune cells that produce proallergic, proinflammatory cytokines (IL-4, IL-5, IL-13), and chemokines (eotaxin) mediate this inflammatory process

- Hypersensitivity or susceptibility to a variety of provocative exposures or triggers can lead to

1. airways inflammation (type 2 and non type 2)
 2. AHR, edema,
 3. basement membrane thickening,
 4. subepithelial collagen deposition,
 5. smooth muscle and mucous gland hypertrophy, and mucus hypersecretion
- all processes that contribute to airflow obstruction

Early Childhood Risk Factors for Persistent asthma

Parental asthma*

Allergy

- Atopic dermatitis (eczema)
- Allergic rhinitis
- Food allergy
- Inhalant allergen sensitization
- Food allergen sensitization

Severe lower respiratory tract infection

- Pneumonia
- Bronchiolitis

Requiring hospitalization

Wheezing apart from colds

Male sex

Low birthweight

Environmental tobacco smoke exposure

Reduced lung function at birth

Formula feeding rather than breastfeeding

09/13/2026

Asthma Triggers

VIRAL INFECTIONS OF THE RESPIRATORY TRACT
AEROALLERGENS IN SENSITIZED ASTHMATIC
PATIENTS

Indoor Allergens

Seasonal Aeroallergens

AIR POLLUTANTS

- Environmental tobacco smoke
- Ozone
- Sulfur dioxide
- Wood- or coal-burning smoke
- Mycotoxins
- Endotoxin
- Dust

STRONG OR NOXIOUS ODORS OR FUMES OCCUPATIONAL EXPOSURES

- Farm and barn exposures
- Formaldehydes, cedar, paint fumes
- Rhinitis
- Sinusitis
- Gastroesophageal reflux

DRUGS

- Aspirin and other nonsteroidal antiinflammatory drugs
- β -Blocking agents

Wheezing phenotypes

Time trend based classification

Not all children who wheeze have asthma.

There are three identified patterns of infant wheezing:

1) **The transient early wheezer (20%):-**

- ✓ Wheeze at least once before the age of 3 yr
- ✓ Never wheeze again

2) **The persistent wheezer (15%)**

- ✓ Has wheezing episodes before age 3 yr
- ✓ Still wheezing at 6 yr of age

3) **The late-onset wheezer (15%)**

- ✓ Does not wheeze before age 3 yr
- ✓ Start wheezing by 6 yr

The remaining 50% of the population never wheezing prior to age 6 yr

Of all the infants who wheezed before 3 years old, almost 60% stopped wheezing by age 6

Types of Childhood Asthma

- Based on natural courses
- **Recurrent wheezing**; in early childhood, primarily triggered by common respiratory viral infections, usually resolves during the preschool/lower school years
- **Chronic asthma**; associated with allergy that persists into later childhood and often adulthood

Asthma is also classified by

Severity Classification*

- Intrinsic disease severity while not taking asthma medications

Intermittent

Persistent:

- Mild
- Moderate
- Severe

Control Classification*

- Clinical assessment while asthma being managed and treated

Well controlled

Not well controlled

Very poorly controlled

Management Patterns

- **Easy-to-control:** well controlled with low levels of daily controller therapy
- **Difficult-to-control:** well controlled with multiple and/or high levels of controller therapies
- **Exacerbators:** despite being well controlled, continue to have severe exacerbations
- **Refractory:** continue to have poorly controlled asthma despite multiple and high levels of controller therapies

AHR, Airways hyperresponsiveness.

Clinical Manifestations

Most common chronic symptoms of asthma

- ✓ Intermittent dry coughing
- ✓ expiratory wheezing

Older children report associated

- ✓ shortness of breath
- ✓ chest tightness
- ✓ Symptoms can be worse at night
- ✓ Daytime symptoms, often linked with physical activities(exercise-induced)

Other asthma symptoms in children can be subtle and nonspecific

Including

- self-impose limitation of physical activities,
- general fatigue (possibly resulting from sleep disturbance), and
- difficulty keeping up with peers in physical activities.
- Chest tightness
- Chest pain
- Limited physical activity
- Improvement of symptoms with asthma medications
- Atopic dermatitis

The presence of risk factors, such as

- history of other allergic conditions
- allergic rhinitis,
- allergic conjunctivitis,
- atopic dermatitis,
- food allergies
- parental asthma, supports the diagnosis of asthma

Physical signs during exacerbations of asthma

- Prolonged expiration with wheezing
- Decreased peripheral breath sounds
- Rhonchi or occasional crackles may be heard due to excess mucus production
- Segmental crackles + decreased breath sounds may indicate asthma-induced atelectasis or lobar pneumonia
- Peak expiratory flow in mild to moderate cases > 40%—

70

severe exacerbations:

- Respiratory distress
- Suprasternal and intercostal retractions and nasal flaring
- Expiratory and inspiratory wheezing
- Markedly decreased breath sounds
- Peak expiratory flow < 40

PEFR

Modified Asthma Predictive Index:

History of ≥ 4 wheezing episodes with at least one physician-diagnosed and either		
One (or more) of the major criteria	or	Two (or more) of the minor criteria
Parental history of asthma		Eosinophilia ($\geq 4\%$)
Skin test positive to aero-allergens		Wheezing unrelated to colds
Eczema (physician-diagnosed atopic dermatitis)		Allergic sensitization to milk, egg, or peanuts

For pre-school-age children with frequent wheezing in the past year, one major criterion or two minor criteria provide a high specificity (97%) and positive predictive value (77%) for persistent asthma into later childhood.

Diagnosis of Asthma in Children ≤ 5 Years (Nelson Textbook of Pediatrics)

Asthma diagnosis in young children is mainly clinical because most children ≤ 5 years cannot perform reliable spirometry.

Three Core Diagnostic Criteria

1. Recurrent Wheezing Episodes

Repeated episodes of wheezing with or without:

Cough

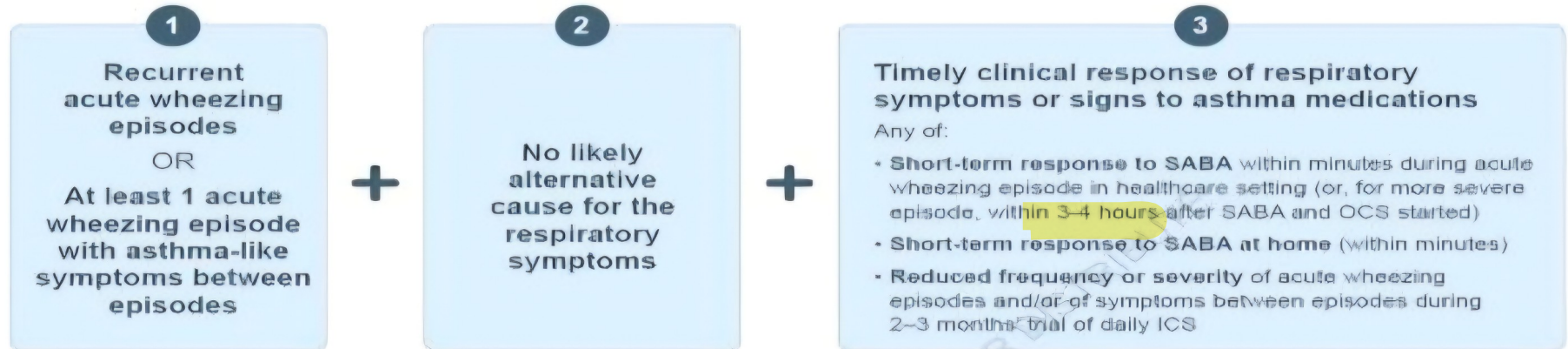
Breathing difficulty

Asthma-like symptoms between episodes

- **2. Exclusion of Alternative Diagnoses**
 - Consider and rule out other causes of recurrent wheeze:
 - Congenital airway abnormalities
 - Foreign body aspiration
 - Cystic fibrosis
 - Other chronic lung diseases
- **3. Response to Asthma Treatment**
 - Diagnosis is supported by improvement after:
 - Short-acting β_2 -agonist (SABA) trial
 - OR
 - 2–3 month trial of daily inhaled corticosteroid (ICS) + as-needed SABA

Diagnosis

Box 10-1. Diagnostic criteria for asthma in children aged 5 years or younger



All three criteria are needed for the diagnosis of asthma in children 5 years and younger

Acute wheezing episode: symptoms such as wheezing on expiration, accessory muscle use, or difficult, fast or heavy breathing, lasting for more than 24 hours

Asthma-like symptoms between episodes (also called interval symptoms): symptoms such as dry cough or wheeze after running, laughing or crying, or during sleep, that occur between acute wheezing episodes

If only 1 or 2 criteria are met, describe as 'suspected asthma', and continue follow-up

A personal or family history of allergic disease may strengthen the diagnosis of asthma, but is not required, and is not specific for asthma

ICS: inhaled corticosteroid; SABA: short-acting beta₂-agonist; OCS: oral; corticosteroid

LUNG FUNCTION TESTS

1. Key Spirometry Parameters

- Spirometry: Plot of airflow vs. time during a rapid, forceful, complete expiration from TLC to RV.
- FVC (Forced Vital Capacity): Max volume exhaled after full inspiration.
- FEV₁ (Forced Expiratory Volume in 1 sec): Single best measure of overall airway function.
- {FEF}_{25-75%}: Mean airflow during middle half of FVC. Sensitive marker for medium and small airway obstruction.

Diagnostic Role of Spirometry in Pediatric Asthma

- 1. Assessing Airway Function & Hyperresponsiveness
- Bronchodilator Reversibility Test:
- Spirometry is performed before and after inhaling a short-acting bronchodilator (e.g., Albuterol).
- A positive response is defined as an improvement in FEV1 of > 12%.
- Bronchial Challenge Test (Provocation):
- Used when baseline spirometry is normal but asthma is suspected.
- A positive test (airway hyperreactivity) is defined as 20% drop in FEV1 (e.g., Methacholine challenge) or 10–15% drop (Exercise challenge).

2. Clinical Significance of Parameters

- FEV1 / FVC Ratio:
- More sensitive measure of impairment in children than FEV1 alone.
- Children often maintain normal FEV1 percent predicted despite significant airflow obstruction, making a reduced ratio ($< 80\text{--}85\%$) key to early detection.
 - (Forced Expiratory Volume in 1 second):
- Essential for assessing current disease severity.
- A reduced baseline FEV1 serves as a strong predictor of risk for future asthma exacerbations.

conti..

- 2. Peak Flow & Airway Inflammation
- **PEFR** (Peak Expiratory Flow Rate): Max flow rate during forced expiration.
- Used to monitor asthma course and treatment response.
- Compare against personal best and predicted values.
- **Diurnal variation > 13%** supports asthma diagnosis.

Clinical application in GINA 2026 guidelines

1) FeNO (Fractional Exhaled Nitric Oxide): Non-invasive marker of eosinophilic/allergic airway inflammation.

>35ppb in children

- Aids in confirming asthma diagnosis and guiding anti-inflammatory therapy.
- Measuring exhaled nitric oxide (FENO):
- a marker of airway inflammation in allergy-associated asthma
- help adjust anti-inflammatory management and confirm the diagnosis of asthma.

conti..

2) Eosinophil count

3) Immunoglobulin E

#The importance of testing bronchodilator responsiveness, bronchial hyperreactivity (e.g., methacholine or exercise challenges), using FEV1/FVC for detecting early/mild airflow obstruction, and using FEV1 to assess risk of future exacerbations are standard concepts

Chest radiograph(CXR)

- The findings of chest radiographs often appear to be normal, it rarely provide information that alters the management of children with acute asthma.
- Indications
- To rule out pneumonia, atelectasis, fever ($>39^{\circ}\text{C}$), presence of focal examination findings(e.g., persistent crackles or decreased breath sounds, crepitus),
- Persistent tachypnea, hypoxemia, or chest pain;
- Severe disease; or
- Uncertainty about the diagnosis

Box 1-2. Criteria for initial diagnosis of asthma in adults (≥18 years) and children (6–17 years)

1. HISTORY OF TYPICAL VARIABLE RESPIRATORY SYMPTOMS

Feature	Symptoms or features that support the diagnosis of asthma
Wheeze, shortness of breath, chest tightness and/or cough (Descriptors may vary by region and by age)	• Symptoms occur variably over time and vary in intensity • Symptoms are often worse at night or on waking • Symptoms are often triggered by exercise, laughter, allergens, cold air • Symptoms worsen after end-exercise (very distinctive) • Symptoms often appear or worsen with viral infections

2. CONFIRMED VARIABLE EXPIRATORY AIRFLOW

Feature	Considerations, definitions, criteria
Excessive variability in expiratory lung function (one or more of the following):	The greater the variations, or the more occasions excess variation is seen, the more confidently the diagnosis of asthma can be made. If spirometry is not possible, PEF may be used, but it is less reliable.
Positive bronchodilator (BD) responsiveness (reversibility) test with spirometry (or PEF[†]) When possible, test during symptoms or in the morning	Measure change 10–15 minutes after 200–400 mcg salbutamol (albuterol) equivalent, compared with pre-BD readings. Positive test more likely if BD test is performed before test: SABA ≥4 hours, long-acting bronchodilators 24–48 hours (see Appendix A). Adults: increase from baseline in FEV ₁ or FVC of ≥12% and ≥200 mL, with confidence if the increase is ≥15% and ≥400 mL; or increase in PEF [†] ≥20% if spirometry is not available Children: increase from baseline in FEV ₁ of ≥12% predicted (or in PEF [†] of ≥20% predicted)
Excessive variability in twice-daily PEF over 2 weeks*	Adults: average daily diurnal PEF variability >10%* Children: average daily diurnal PEF variability >13%*
Increase in lung function after 4 weeks of ICS-containing treatment	Adults: increase from baseline in FEV ₁ by ≥12% and ≥200 mL (or PEF [†] by ≥12% and ≥200 mL) after 4 weeks of daily ICS-containing treatment Children: increase from baseline in FEV ₁ of ≥12% predicted (or in PEF [†] of ≥12% predicted)
Positive bronchial provocation test	Adults: Fall from baseline in FEV ₁ of ≥20% with standard doses of methacholine or ≥15% with standardized hyperventilation, hypertonic saline or mannitol challenge Children: fall from baseline in FEV ₁ of >12% predicted (or fall in PEF [†] >15% predicted) with standardized exercise challenge. If FEV ₁ decreases during a challenge test, check that FEV ₁ /FVC ratio has not decreased, since incomplete inhalation, e.g., due to inducible laryngeal obstruction or poor effort, can result in a false reduction in FEV ₁ .
Excessive variation in lung function between visits (good specificity but poor sensitivity)	Adults: variation in FEV ₁ of ≥12% and ≥200 mL (or in PEF [†] of ≥20%) between visits Children: variation of ≥12% in FEV ₁ (or ≥15% in PEF [†]) between visits

ROLE OF TYPE 2 BIOMARKERS IN DIAGNOSIS OF ASTHMA

In patients with typical asthma symptoms, if spirometry or PEF is not available or testing is negative, elevated FeNO (adults/adolescents: >50 ppb; children: >35 ppb) or blood eosinophils above national/regional reference range can support the diagnosis of Type 2 asthma, but can also be due to non-asthma conditions. Lower levels of FeNO or blood eosinophils do not rule out asthma. FeNO and blood eosinophils vary substantially by sex, age and (for FeNO) device and season. Eosinophils are higher in the morning, and FeNO is lower in the morning. See Appendix A for more details (p.21).

OTHERs

- CT Scan (Computed Tomography): Used to further define comorbidities (e.g., allergic bronchopulmonary aspergillosis, severe sinus disease), differential diagnoses, and anatomical/structural complications.
- Skin Prick Testing: Used to assess allergic sensitization (atopic status) to specific aeroallergens.
- Genetic Testing: Indicated specifically to exclude monogenic obstructive lung diseases that mimic asthma (such as Cystic Fibrosis or Primary Ciliary Dyskinesia), particularly in patients presenting with recurrent bronchitis or chronic productive cough.

Age	Common	Uncommon	Rare
Less than 6 months	Bronchiolitis Gastro-esophageal reflux	Aspiration pneumonia Bronchopulmonary dysplasia Congestive heart failure Cystic fibrosis	Asthma Foreign body aspiration
6 months - 2 years	Bronchiolitis Foreign body aspiration	Aspiration pneumonia Asthma Bronchopulmonary dysplasia Cystic fibrosis Gastro-esophageal reflux	Congestive heart failure
2 - 5 years	Asthma Foreign body aspiration	Cystic fibrosis Gastro-esophageal reflux Viral pneumonia	Aspiration pneumonia Bronchiolitis Congestive heart failure Gastro-esophageal reflux

Differential Diagnosis of Childhood Asthma

UPPER RESPIRATORY TRACT CONDITIONS

Allergic rhinitis*
Chronic rhinitis*
Sinusitis*
Adenoidal or tonsillar hypertrophy
Nasal foreign body

MIDDLE RESPIRATORY TRACT CONDITIONS

Laryngotracheobronchomalacia*
Laryngotracheobronchitis (e.g., pertussis)*
Laryngeal web, cyst, or stenosis
Exercise-induced laryngeal obstruction
Vocal cord dysfunction*
Vocal cord paralysis
Tracheoesophageal fistula
Vascular ring, sling, or external mass compressing on the airway (e.g., tumor)
Foreign body aspiration*
Chronic bronchitis from environmental tobacco smoke exposure*
Toxic inhalations

LOWER RESPIRATORY TRACT CONDITIONS

Bronchopulmonary dysplasia (chronic lung disease of preterm infants)
Viral bronchiolitis*
Gastroesophageal reflux*
Causes of bronchiectasis:
Cystic fibrosis
Immune deficiency
Allergic bronchopulmonary mycoses (e.g., aspergillosis)
Chronic aspiration
Immotile cilia syndrome, primary ciliary dyskinesia
Bronchiolitis obliterans
Interstitial lung diseases
Hypersensitivity pneumonitis
Pulmonary eosinophilia, Churg-Strauss vasculitis
Pulmonary hemosiderosis
Tuberculosis
Pneumonia
Pulmonary edema (e.g., congestive heart failure)
Medications associated with chronic cough:
Acetylcholinesterase inhibitors
 β -Adrenergic antagonists
Angiotensin-converting enzyme inhibitors

Box 10-4. Common differential diagnoses of asthma in children 5 years and younger

If the symptoms or signs below are present, consider...	Condition
Mainly cough and runny congested nose for <10 days, <i>without</i> wheezing or difficulty breathing	Viral upper respiratory tract infection
Cough when feeding, recurrent chest infections	Gastroesophageal reflux +/- pharyngeal dysphagia
Sudden onset of symptoms, unilateral wheeze	Inhaled foreign body Other conditions including tuberculosis
Protracted paroxysms of coughing, often with stridor and vomiting	Pertussis
Persistent wet cough	Protracted bacterial bronchitis Tuberculosis
Noisy breathing when crying or eating; harsh cough	Tracheomalacia
Cardiac murmurs, failure to thrive	Congenital heart disease
Pre-term delivery, symptoms since birth	Bronchopulmonary dysplasia
Excessive cough and mucus production, gastrointestinal symptoms, failure to thrive	Cystic fibrosis
Cough and recurrent chest infections; neonatal respiratory distress, chronic ear infections and persistent nasal discharge from birth	Primary ciliary dyskinesia
Noisy breathing, feeding difficulties	Vascular ring

Treatment

- Management of asthma should have the following components:
- assessment and monitoring of disease activity;
- education to enhance patient and family knowledge and skills for self-management;
- identification and management of precipitating factors and comorbid conditions that worsen asthma; and
- appropriate selection of medications to address the patient's needs.

GINA 2026

1. **SABA-only therapy is strongly discouraged.** ✓
2. **Preferred reliever = Low-dose ICS-formoterol.**
3. **Oxygen only if $\text{SpO}_2 < 92\%$.**
4. **Excess SABA may cause lactic acidosis.**
5. **ICS-formoterol expanded into mild exacerbation pathways.**
6. **AIR therapy added in pediatric asthma (5–15 years).**
7. **MART/SMART therapy remains preferred in moderate-severe asthma.**
8. **Depemokimab added for severe eosinophilic asthma.**
9. **Precision medicine and biomarker-guided therapy emphasized.**
10. **GINA 2026 focuses on anti-inflammatory, biomarker-driven asthma management.**

Recurrent/chronic cough, wheeze, chest tightness and/or shortness of breath

Diagnosis

- Symptoms
- Exacerbations
- Risk factors
- Triggers
- Lung function
- Differential diagnosis

Asthma

Management

• *Assessment and monitoring*

- Assess severity
- Monitor control
- Med adverse effects

• *Education*

- Key elements

• *Control environmental factors and co-morbid conditions*

- Environmental controls
- Co-morbidities

• *Medications*

- Long-term controllers
- Quick relievers

• *Exacerbations*

- Assessment
- Management
- High-risk features
- Home asthma action plan

Optimal goal: Well controlled asthma

• *Reduce impairment*

- Prevent chronic symptoms
- Prevent sleep disturbance
- Infrequent SABA need
- Maintain (near) normal lung function
- Maintain normal activity

• *Reduce risk*

- Prevent exacerbations
- Reduce exacerbation severity/duration
- Prevent reduced lung growth
- No (minimal) adverse effects of therapy

Pediatric Asthma — Regular Assessment & Monitoring

- 1. Key Concepts
- Asthma Control: The degree to which manifestations of asthma are minimized by therapeutic interventions and treatment goals are met.
- Asthma Severity (Retrospective Assessment): Assessed from the level of treatment required to maintain good symptom control and prevent exacerbations (e.g., Mild = controlled with low-dose ICS; Severe = requires high-dose ICS-LABA or biologics).

2. The Two Assessment Domains

- A. Impairment (Current Status)
- Symptom Frequency: Day and night symptom burden over the past 4 weeks.
- Activity Limitation: Impact on daily activities, exercise, play, and school attendance.
- Lung Function: Objective spirometry (FEV1, FEV1/FVC) or PEF monitoring.
- B. Risk (Future Outcomes)
- Exacerbations: History of severe attacks requiring oral systemic corticosteroids or emergency visits.
- Airway & Lung Growth: Risk of progressive loss of lung function or reduced lung growth in children.
- Reliever Overuse: Using ≥ 3 SABA canisters/year increases risk of severe exacerbations and asthma-related mortality.
- Medication Side Effects: Monitoring for adverse effects of long-term controller therapy (e.g., growth velocity checks with ICS).

Table 144-8**Assessing Asthma Severity and Initiating Treatment for Patients Who Are Not Currently Taking Long-Term Control Medications***

CLASSIFICATION OF ASTHMA SEVERITY				
	Intermittent	PERSISTENT		
		Mild	Moderate	Severe
COMPONENTS OF SEVERITY				
Impairment				
Daytime symptoms	≤2 days/wk	>2 days/wk but not daily	Daily	Throughout the day
Nighttime awakenings:				
Age 0-4 yr	0	1-2x/mo	3-4x/mo	>1x/wk
Age ≥5 yr	≤2x/mo	3-4x/mo	>1x/wk but not nightly	Often 7x/wk
Short-acting β ₂ -agonist use for symptoms (not for prevention of exercise-induced bronchospasm)	≤2 days/wk	>2 days/wk but not daily, and not more than 1x on any day	Daily	Several times per day
Interference with normal activity	None	Minor limitation	Some limitation	Extreme limitation
Lung function:				
FEV ₁ % predicted, age ≥5 yr	Normal FEV ₁ between exacerbations >80% predicted	≥80% predicted	60-80% predicted	<60% predicted
FEV ₁ :FVC ratio [†] :				
Age 5-11 yr	>85%	>80%	75-80%	<75%
Age ≥12 yr	Normal	Normal	Reduced 5%	Reduced >5%
Risk				
Exacerbations requiring systemic corticosteroids:				
Age 0-4 yr	0-1/yr (see notes)	≥2 exacerbations in 6 mo requiring systemic corticosteroids or ≥4 wheezing episodes/yr lasting >1 day and risk factors for persistent asthma		
Age ≥ 5 yr	0-1/yr (see notes)	≥2/yr (see notes)	≥2/yr (see notes)	≥2/yr (see notes)
Consider severity and interval since last exacerbation.				
Frequency and severity may fluctuate over time for patients in any severity category.				
Relative annual risk of exacerbations may be related to FEV ₁ .				

Table 144-9 Assessing Asthma Control and Adjusting Therapy in Children*

	CLASSIFICATION OF ASTHMA CONTROL		
	Well-Controlled	Not Well-Controlled	Very Poorly Controlled
COMPONENTS OF CONTROL			
Impairment			
Symptoms	≤2 days/wk but not more than once on each day	>2 days/wk or multiple times on ≤2 days/wk	Throughout the day
Nighttime awakenings:			
Age 0-4 yr	≤1x/mo	>1x/mo	>1x/wk
Age 5-11 yr	≤1x/mo	≥2x/mo	≥2x/wk
Age ≥12 yr	≤2x/mo	1-3x/wk	≥4x/wk
Short-acting β ₂ -agonist use for symptoms (not for exercise-induced bronchospasm pretreatment)	≤2 days/wk	>2 days/wk	Several times per day
Interference with normal activity	None	Some limitation	Extremely limited
Lung function:			
Age 5-11 yr:			
FEV ₁ (% predicted or peak flow)	>80% predicted or personal best	60-80% predicted or personal best	<60% predicted or personal best
FEV ₁ /FVC:	>80%	75-80%	<75%
Age ≥ 12 yr:			
FEV ₁ (% predicted or peak flow)	>80% predicted or personal best	60-80% predicted or personal best	<60% predicted or personal best
Validated questionnaires[†]:			
Age ≥ 12 yr:			
ATAQ	0	1-2	3-4
ACQ	≤0.75	≤1.5	N/A
ACT	≥20	16-19	≤15
Risk			
Exacerbations requiring systemic corticosteroids:			
Age 0-4 yr	0-1/yr	2-3/yr	>3/yr
Age ≥5 yr	0-1/yr	≥2/yr (see notes)	
Consider severity and interval since last exacerbation.			
Treatment-related adverse effects	Medication side effects can vary in intensity from none to very troublesome and worrisome. The level of intensity does not correlate to specific levels of control but should be considered in the overall assessment of risk.		
Reduction in lung growth or progressive loss of lung function	Evaluation requires long-term follow-up care.		

Component 2: Patient Education

Table 185.11

Key Elements of Productive Clinic Visits for Asthma

Standardize assessment of asthma control (e.g., Asthma Control Test, exacerbations in past 12 mo)

Specify goals of asthma management

Explain basic facts about asthma

- Contrast normal vs asthmatic airways
- Link airways inflammation, “twitchiness,” and bronchoconstriction
- Long-term-control and quick-relief medications
- Address concerns about potential adverse effects of asthma pharmacotherapy

Teach, demonstrate, and have patient show proper technique

- Inhaled medication use (spacer use with metered-dose inhaler)

Investigate and manage factors that contribute to asthma severity

- Environmental exposures
- Comorbid conditions

Create written two-part Asthma Action Plan (see Fig. 185.5)

- Daily management
- Action plan for asthma exacerbations

Regular follow-up visits

- Twice yearly (more often if asthma not well controlled)
- Monitor lung function at least annually

Asthma Action Plan

Asthma Action Plan

For: _____ Doctor: _____ Date: _____

Doctor's Phone Number _____ Hospital/Emergency Department Phone Number _____

GREEN ZONE

Doing Well

- No cough, wheeze, chest tightness, or shortness of breath during the day or night
- Can do usual activities

And, if a peak flow meter is used,

Peak flow: more than _____
(80 percent or more of my best peak flow)

My best peak flow is: _____

Take these long-term-control medicines each day (Include an anti-inflammatory).

Medicine

How much to take

When to take it

Identify and avoid and control the things that make your asthma worse, like (list here):

Before exercise, if prescribed, take: ☐ 2 or ☐ 4 puffs _____ 5 to 60 minutes before exercise

YELLOW ZONE

Asthma Is Getting Worse

- Cough, wheeze, chest tightness, or shortness of breath, or
- Waking at night due to asthma, or
- Can do some, but not all, usual activities

-Or-

Peak flow: _____ to _____
(50 to 79 percent of my best peak flow)



Add: quick-relief medicine—and keep taking your GREEN ZONE medicine.

_____ ☐ 2 or ☐ 4 puffs, every 20 minutes for up to 1 hour
(short-acting beta₂-agonist) ☐ Nebulizer, once

If applicable, remove yourself from the thing that made your asthma worse.

If your symptoms (and peak flow, if used) return to GREEN ZONE after 1 hour of above treatment:



☐ Continue monitoring to be sure you stay in the green zone.

-Or-

If your symptoms (and peak flow, if used) do not return to GREEN ZONE after 1 hour of above treatment:

☐ Take: _____ ☐ 2 or ☐ 4 puffs or ☐ Nebulizer

_____ (short-acting beta₂-agonist)

☐ Add: _____ mg per day. For _____ (3–10) days

_____ (oral corticosteroid)

☐ Call the doctor _____ (phone) ☐ before/ ☐ within _____ hours after taking the oral corticosteroid.

RED ZONE

Medical Alert!

- Very short of breath, or
- Quick-relief medicines have not helped, or
- Cannot do usual activities, or
- Symptoms are same or get worse after 24 hours in Yellow Zone.

-Or-

Peak flow: less than _____
(50 percent of my best peak flow)

Take this medicine:

☐ _____ ☐ 4 or ☐ 6 puffs or ☐ Nebulizer
(short-acting beta₂-agonist)

☐ _____ mg
(oral corticosteroid)

Then call your doctor NOW. Go to the hospital or call an ambulance if

■ You are still in the red zone after 15 minutes AND

■ You have not reached your doctor.

DANGER SIGNS

- Trouble walking and talking due to shortness of breath
- Lips or fingernails are blue



- Take ☐ 4 or ☐ 6 puffs of your quick-relief medicine AND
- Go to the hospital or call for an ambulance _____ NOW!

Component 3: Control of Factors Contributing to Poor Asthma Control

Table 185.12

Control of Factors Contributing to Asthma Severity

ELIMINATE OR REDUCE PROBLEMATIC ENVIRONMENTAL EXPOSURES

Environmental tobacco smoke elimination or reduction in home and automobiles

Allergen exposure elimination or reduction in sensitized asthmatic patients

- Pets (cats, dogs, rodents, birds)
- Pests (mice, rats)
- Dust mites
- Cockroaches
- Molds

Other airway irritants

- Wood- or coal-burning smoke
- Strong chemical odors and perfumes (e.g., household cleaners)
- Dusts

TREAT COMORBID CONDITIONS

Rhinitis

Sinusitis

Gastroesophageal reflux

Component 4: Principles of Asthma Pharmacotherapy

Long-Term **Controller** Medications

- ❖ Inhaled Corticosteroids
- ❖ Long-Acting Inhaled β -Agonists
- ❖ Leukotriene-Modifying Agents
- ❖ Nonsteroidal Anti-inflammatory Agents
- ❖ Theophylline
- ❖ Anti-Immunoglobulin E (Omalizumab)

Quick-**Reliever** Medications

- ❖ Short-Acting Inhaled β -Agonists
- ❖ Anticholinergic Agents
- ❖ Systemic Corticosteroids

Inhaled Corticosteroids ICS therapy

- serves as the cornerstone of therapy in persistent asthma
- as it improves lung function; reduces asthma symptoms, AHR, and use of “rescue” medications; improves quality of life; and
- most importantly reduces exacerbations, the need for systemic corticosteroids, urgent care visits, and hospitalizations.
- Epidemiologic studies have also shown that ICS therapy in adults substantially lowers the risk of death attributable to asthma if used regularly

Cont.....

- ICSs are available in metered-dose inhalers (MDIs), in dry powder inhalers (DPIs), or in suspension for nebulization.
- The risk of adverse effects from ICS therapy is related to the dose and frequency of administration
- High doses ($\geq 1,000$ $\mu\text{g/day}$ in children) and frequent administration (4 times per day) are more likely to have both local and systemic adverse effects.

Cont.....

- The most common ICS adverse effects are local: **oral candidiasis (thrush) and dysphonia (hoarse voice)**.
- The incidence of these local effects can be greatly minimized by using a spacer with an MDI with the ICS
- Mouth rinsing using a “swish and spit” technique after ICS use is also recommended.
- The potential for growth suppression and osteoporosis with longterm ICS use had been an unanswered concern.

Long-Acting Inhaled β -Agonists

- LABAs are considered to be daily controller medications, (salmeterol, formoterol) ***not intended for use as rescue medication for acute asthma symptoms or exacerbations, nor as monotherapy for persistent asthma.***
- Given their long duration of action (≥ 12 hours), they are well suited for patients with nocturnal asthma and for individuals who require frequent SABA use during the day to prevent exercise-induced bronchospasm.
- Improve asthma control and reduce exacerbations when added to ICS

Combination ICS/LABA Therapy

- is recommended for patients who are suboptimally controlled with ICS therapy alone and those with moderate or severe persistent asthma.
- In most patients who are inadequately controlled with ICS alone, combination ICS/LABA therapy is superior to add-on therapy with either an LTRA or LAMA or doubling the ICS dose.
- Benefits include improvement in baseline lung function, less need for rescue SABA therapy, improved quality of life, and fewer asthma exacerbations

Cont.....

- large randomized trials (>23,000 adults/adolescents and >6,000 children aged 4–11) comparing ICS alone vs. combination ICS/LABA.
- **Key Finding:** ICS/LABA combination therapy does NOT increase the risk of asthma-related hospitalizations, intubations, or deaths compared to ICS alone when given in a single device.
- In children uncontrolled on low-dose ICS, adding a LABA was almost twice as likely to succeed as doubling the ICS dose or adding Montelukast.

Leukotriene-Modifying Agents

- **Leukotrienes are potent proinflammatory mediators that can induce bronchospasm, mucus secretion, and airways edema.**
- Two classes of leukotriene modifiers have been developed: inhibitors of leukotriene synthesis and LTRAs.
- **LTRAs have bronchodilator and anti-inflammatory properties and reduce exercise-, aspirin-, and allergen-induced bronchoconstriction.**
- *LTRAs are recommended as alternative treatment for mild persistent asthma and as add-on medication with ICS for moderate persistent asthma.*

Cont.....

- LTRAs are less effective than ICS for mild persistent asthma (ICS improves baseline lung function by 5–15%, compared to 2.5–7.5% for LTRAs).

Dosing Frequency

- **Montelukast:** Approved for children ≥ 1 year old (Given once daily).
- **Zafirlukast:** Approved for children ≥ 5 years old (Given twice daily).

Neuropsychiatric Risk: Risk of serious behavioral and mood-related changes (e.g., agitation, depression).

Families must be counseled to **discontinue Montelukast immediately** and contact their physician if mood or behavioral changes occur.

Nonsteroidal Anti-inflammatory Agents

- Cromolyn and nedocromil are NSAID agents that can **inhibit allergen-induced asthmatic responses and reduce exercise-induced bronchospasm.**
- Both drugs are considered alternative anti-inflammatory drugs for children with mild persistent asthma.
- while cromolyn and nedocromil were historically used as alternative controllers, they are no longer widely available or routinely recommended.

Cont.....

- They are significantly less effective than low-dose Inhaled Corticosteroids (ICS) in controlling symptoms, improving lung function, and preventing exacerbations.
- They require frequent dosing (3–4 times daily), leading to poor adherence compared to once- or twice-daily ICS inhalers.

Theophylline

- In addition to its bronchodilator effects, it has anti-inflammatory properties as a phosphodiesterase inhibitor.
- When used long-term it can reduce asthma symptoms and the need for rescue SABA use.
- It can be used as an alternative add-on/adjunctive controller for persistent asthma (used alongside an ICS) when preferred combination therapies (like LABAs or LTRAs) are unavailable or unaffordable.
- It is no longer considered a first-line agent for young children, in whom there is significant variability in the absorption and metabolism of different theophylline preparations, necessitating frequent dose monitoring (drug blood levels) and adjustments.

Others

- **Long-Acting Inhaled Anticholinergics**

- ✓ Tiotropium

Allergen Immunotherapy

Biologic Therapies

- ✓ Omalizumab (Anti-IgE Antibody)
- ✓ Mepolizumab and Reslizumab (Anti-IL-5 Antibodies)
- ✓ Dupilumab (Anti-IL-4 Receptor α Antibody)

Short-Acting Inhaled β -Agonists

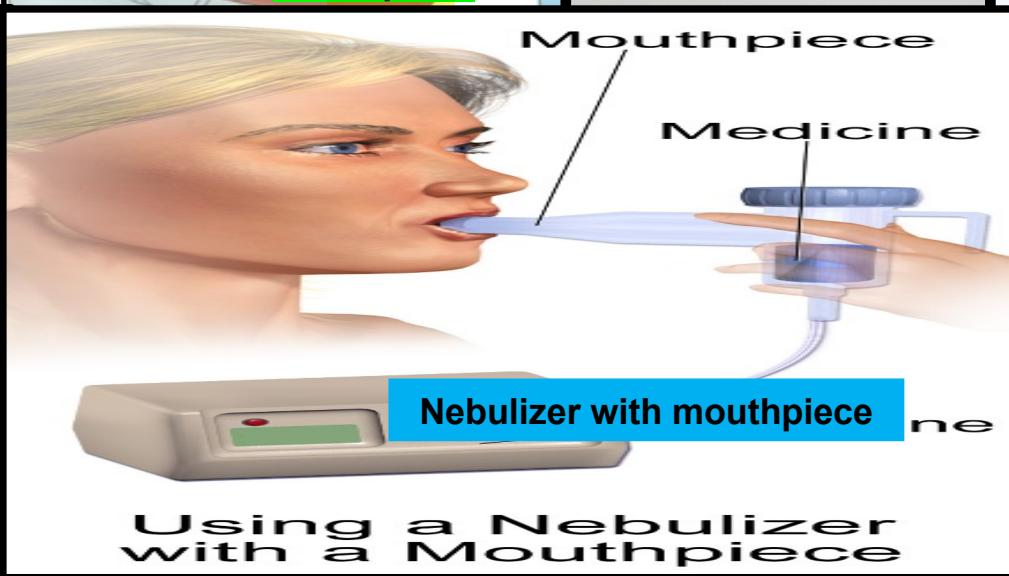
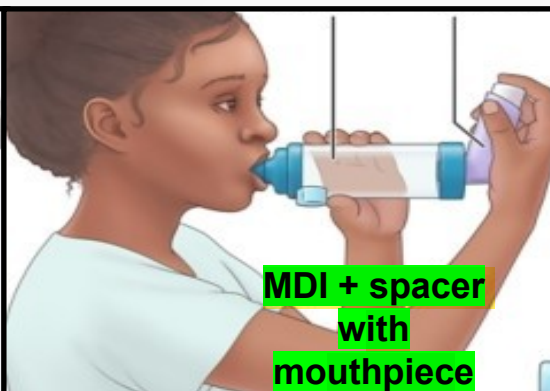
- Are the drug of choice for acute asthma symptoms (as a rescue medication) and for preventing exercise induced bronchospasm.
- **β -Agonists cause bronchodilation by**
 - inducing airway smooth muscle relaxation
 - reducing vascular permeability and airways edema
 - improving mucociliary clearance.
 - reversing & preventing contraction of airway smooth-muscle cells by known bronchoconstrictors
- **No effects on inflammatory cells in the airways and there is no reduction in AHR.**
- increased use of rescue SABA {means: ≥ 1 MDI per month or ≥ 3 MDIs per year (200 inhalations/MDI) } reflects poor asthma control, which is a risk factor for asthma death

Anticholinergic Agents

- **prevent cholinergic nerve-induced bronchoconstriction and mucus secretion.**
- Are less potent than the β -agonists.
- **Inhaled ipratropium is used primarily in the treatment of acute severe asthma.**
- When used in combination with albuterol, ipratropium can improve lung function and reduce the rate of hospitalization in children who present to the emergency department with acute asthma.
- Side effects are not usually a problem because there is **little or no systemic absorption.**

Delivery Devices and Inhalation Technique

- Inhaled medications are delivered in aerosolized form in an MDI, as a DPI formulation, or in a suspension form delivered via a nebulizer.
- Spacer devices, recommended for the administration of all MDI medications, are simple and inexpensive tools that
 - 1) decrease the coordination required to use MDIs, especially in young children;
 - 2) improve the delivery of inhaled drug to the lower airways; and
 - 3) minimize the risk of drug and propellant-mediated oropharyngeal adverse effects (dysphonia and thrush).



Asthma Medication Terms

- **Maintenance Treatment:** Medication prescribed for daily, regular schedule use.
- **Controller:** Medication that targets both symptom control and future risk reduction.
- **Reliever:** An inhaler used as needed for quick symptom relief.
- **Anti-Inflammatory Reliever (AIR):** A reliever inhaler combining a low-dose ICS with a rapid-acting bronchodilator.
- **Maintenance-and-Reliever Therapy (MART):** A regimen using a single ICS-Formoterol inhaler for both daily maintenance and as-needed relief

Asthma Treatment Steps for Children Aged 5 Years and Younger

- Stepwise Approach to adjust treatment up or down to achieve symptom control while minimizing side effects and exacerbation risks.

Step 1: Infrequent Wheezing / Mild Symptoms

- Preferred Option: As-needed inhaled SABA.
- Indicated for children with symptoms ≤ 2 times per week over 1 month period.

Step 2: Preferred Initial Controller Therapy

- Indication: Symptoms > 2 times per week over a 1 month period, or ≥ 1 severe exacerbation in the past year.
- Preferred Option: Daily low-dose ICS plus as-needed SABA (trial for at least 2–3 months).

Step 3: Inadequate Control on Low-Dose ICS

- Check First: Before stepping up, verify asthma diagnosis, inhaler technique, treatment adherence, and environmental exposures (e.g., tobacco smoke).
- Preferred Option: Double the "low" daily ICS dose (Medium-dose ICS) plus as-needed SABA.

Step 4: Severe / Persistent Uncontrolled Asthma

- Primary Requirement: Refer for expert/specialist assessment.
- Interim Care: Continue controller treatment (medium-dose ICS + as-needed SABA) while re-evaluating triggers and diagnosis.

Specialist Options (Limited Evidence):

- Add-on LAMA (e.g., Tiotropium).
- Add-on LTRA to maintenance ICS.
- Combination ICS-LABA (insufficient data in children < 4 years).
- **No MART:** Maintenance-and-Reliever Therapy (MART) with ICS-Formoterol has no published safety data for children < 5 years.

Treatment tracks for adults and adolescents and children 6-11 years

Treatment figures for adults and adolescents shows two tracks, largely based on choice of reliever.

Track 1: Preferred Approach (ICS-Formoterol Reliever)

- Reliever: As-needed low-dose ICS-Formoterol.

Dosing Protocol (MART):

- Steps 1–2: As-needed ICS-Formoterol alone (no daily controller).
- Steps 3–5: Daily ICS-Formoterol maintenance + as-needed ICS-Formoterol reliever.

Why Preferred:

- Cuts severe exacerbation risk vs. SABA relievers (with equal symptom control).
- Uses a single inhaler for both maintenance and relief (Steps 1–4).
- **Key Restriction:** Do not use as a reliever if on a different ICS-LABA maintenance inhaler (e.g., Fluticasone-Salmeterol).

Track 2: Alternative Approach (ICS-SABA or SABA Reliever)

- Reliever: As-needed ICS-SABA combination or SABA alone.
- Indications: Use only if Track 1 is unavailable, or if the patient is stable, adherent, and exacerbation-free for ≥ 1 year.

Dosing Protocol:

- Step 1: As-needed SABA + ICS taken together (or combination ICS-SABA).
- Steps 2–5: Regular daily maintenance ICS (or ICS-LABA) + separate as-needed SABA (or ICS-SABA).
- **Main Risk:** ⚠ Poor adherence to daily maintenance ICS leaves the patient relying solely on SABA, significantly raising exacerbation risk.

Pediatric Considerations (Children 6–11 Years)

- **AIR Regimens (Steps 1–2):** Use as-needed low-dose ICS-Formoterol or ICS-SABA to eliminate SABA-only risks.

Step-Up Pathways

- :As-needed ICS-Formoterol---> Step up to low-dose ICS-Formoterol MART.
- As-needed ICS-SABA -----> Step up to daily ICS + as-needed ICS-SABA.

General Step-Up & Step-Down Rules

Before Stepping Up: Always verify inhaler technique, adherence, allergen exposures, and comorbidities first.

Stepping Down: Once control is stable for 2–3 months, gradually step down to find the minimum effective dose.

Rule: ✗ Never stop ICS completely unless the asthma diagnosis is revised.

ACUTE EXACERBATIONS OF ASTHMA

- An acute or subacute worsening of symptoms (cough, wheeze, shortness of breath) with a drop in lung function (PEF/FEV1 in children ≥ 6 years) or increased work of breathing (retractions, tachypnea in children ≤ 5 years) that requires a escalation in therapy.
- A severe exacerbation of asthma that does not improve with standard therapy is termed **status asthmaticus**.
- Exacerbations usually occur in response to exposure to an external agent/triggers

Cont....

- In infants and children, most cardiac arrests result from progressive respiratory failure and/or shock
- Hence rapid intervention and evaluation is mandatory
- The evaluation includes:
 - General assessment via the pediatric assessment triangle (brief visual and auditory observation of child's overall appearance, work of breathing, circulation)
 - Primary assessment (rapid evaluation of cardiopulmonary and neurologic function)
 - Secondary assessment

Risk Factors for Asthma Exacerbations

- **Primary Driving Factor**

Poor Asthma Symptom Control: Substantially increases overall exacerbation risk.

Asthma symptom control is defined by how much asthma impairs the patient's day-to-day life over a specific recall period (typically the **past 4 weeks**).

Asthma Symptom Control Tool

Ask the patient\care giver if they have experienced any of the following in the **past 4 weeks**:

- **Daytime asthma symptoms more than twice a week?**
- **Any night waking** due to asthma?
- **SABA reliever** needed for symptoms more than twice a week?
(Excluding pre-exercise use)
- Any activity limitation due to asthma?

- Based on how many "YES" answers the patient has:

Classification	Number of "YES" Answers	Clinical Status
Well Controlled	0	Target clinical state.
Partly Controlled	1 – 2	Poor / Suboptimal Control ->Increased exacerbation risk.
Uncontrolled	3 – 4	Severe Poor Control -> High immediate risk of exacerbation.

We say a patient has Poor Asthma Symptom Control whenever they answer YES to 1 or more of this criteria.

A state of frequent symptoms, high reliever reliance, or activity limits (past 4 weeks) that directly signals active airway inflammation and predicts a high risk of future exacerbations.

Independent & Modifiable Risk Factors ("Treatable Traits")

- **Clinical History:** History of ≥ 1 exacerbation in the previous year.
- **Medication & Compliance Issues:**
 - ✓ Poor ICS adherence or incorrect inhaler technique.
 - ✓ SABA Overuse: Prescribing ≥ 3 canisters/year (≥ 200 doses each) increases severe exacerbation and mortality risk.
- **Objective Measures:** Low baseline lung function (FEV1), raised FeNO, or elevated blood eosinophil count.
- **Comorbidities:** Chronic rhinosinusitis and comorbid allergic disease.
- **Exposures & Demographics:** Smoking, air pollution, weather changes, respiratory viruses, aeroallergens, and poverty.

- Initial management supports airway, breathing, and circulation:
- Airway – Key steps in basic airway management include
 - Provide 100 percent inspired oxygen
 - Allow child to assume position of comfort or manually open airway
 - Clear airway (suction)
 - Insert oropharyngeal airway or nasopharyngeal airway if consciousness impaired
- Breathing – The clinician should assist ventilation manually in patients not responding to basic airway maneuvers, monitor oxygenation by pulse oximetry.
- Circulation – Key interventions consist of monitoring heart rate and rhythm and establishing vascular access& fluid administration.

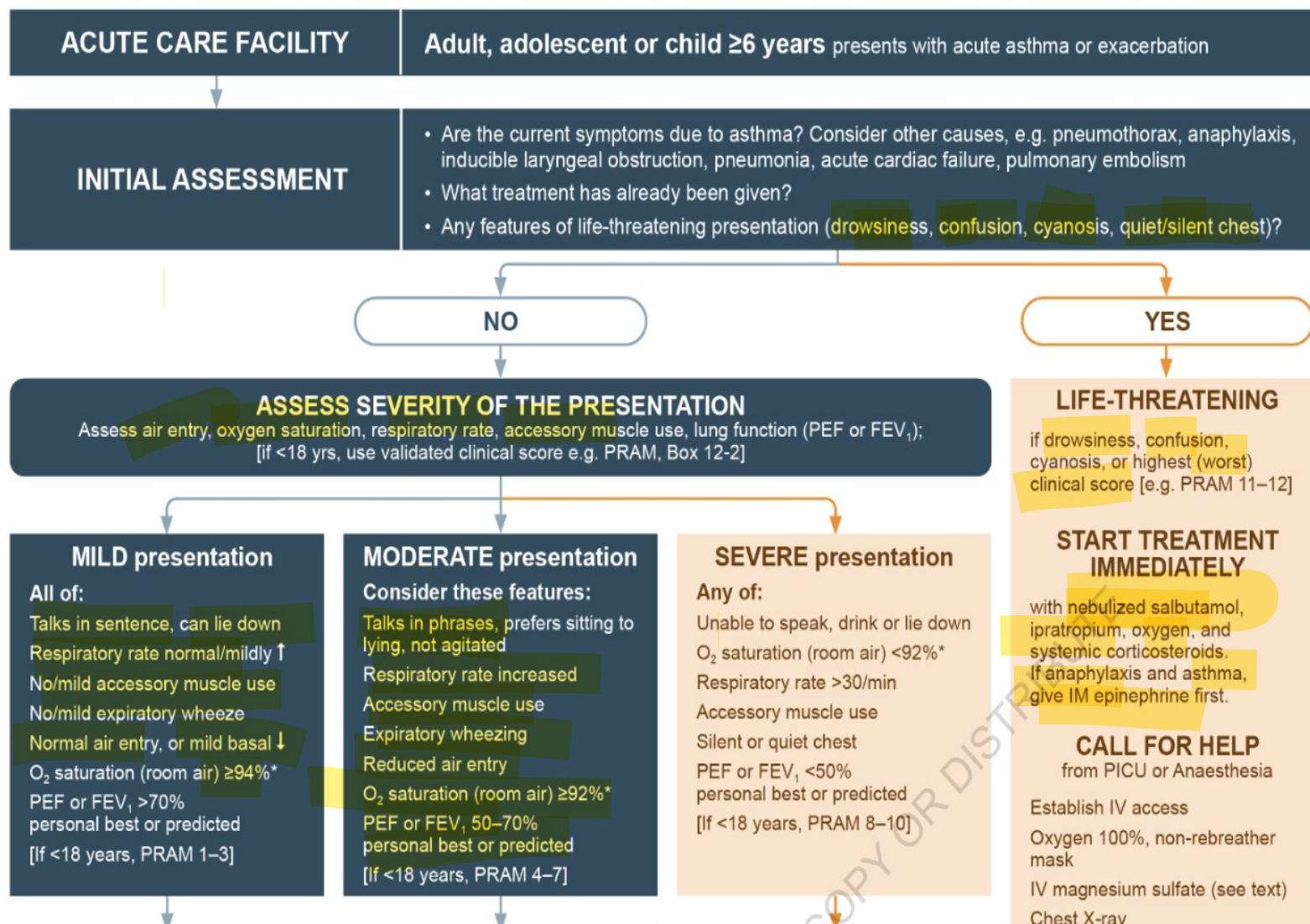
Identifying patients at risk of asthma-related death

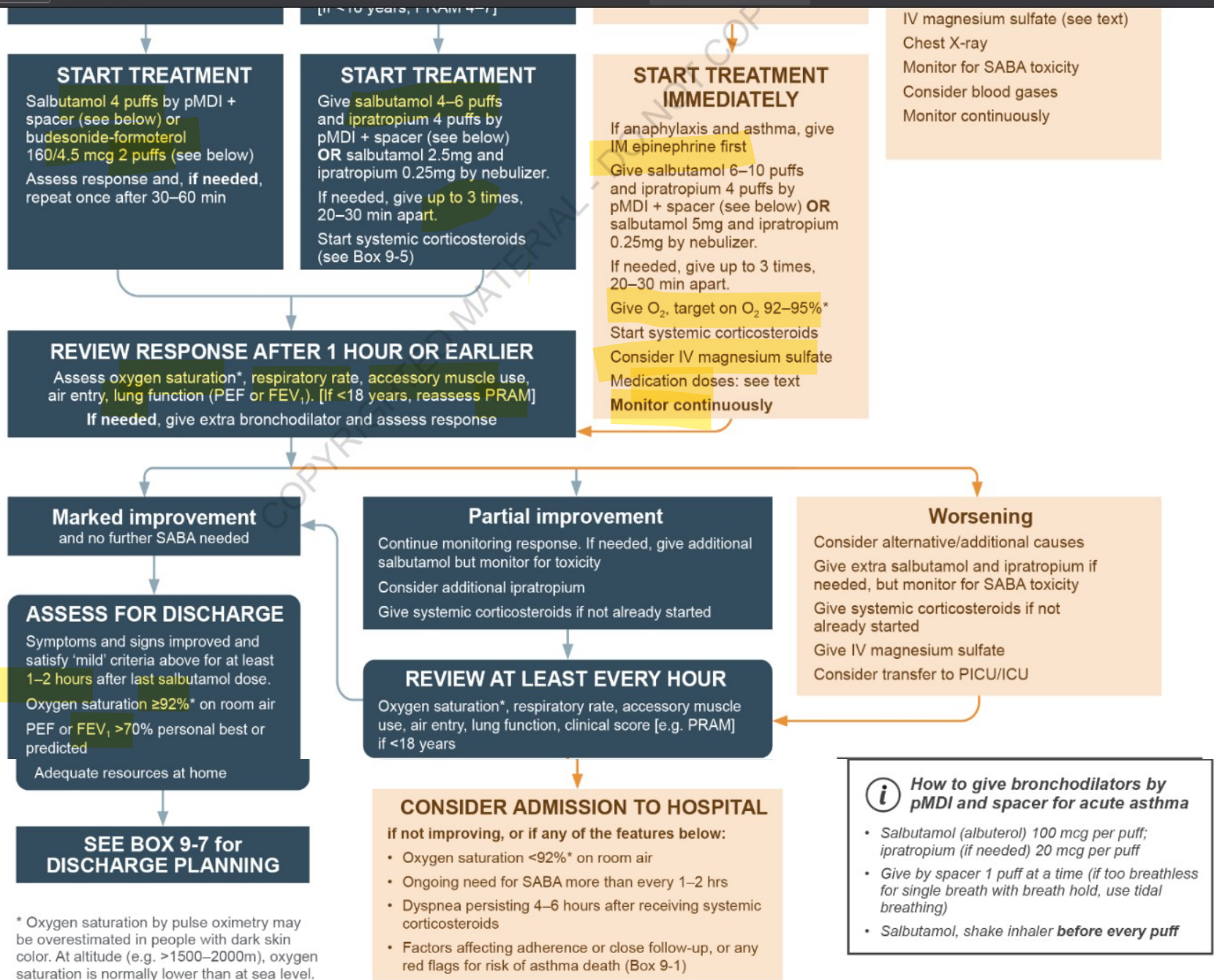
- A history of near-fatal asthma requiring intubation and mechanical ventilation
- Hospitalization or emergency care visit for asthma in the past year
- Currently using or having recently stopped using oral corticosteroids (a marker of event severity)
- Not currently using inhaled corticosteroids
- Over-use of SABAs, especially use of more than one canister of salbutamol (or equivalent) monthly
- Poor adherence with ICS-containing medications
- A history of psychiatric disease or psychosocial problems
- Food allergy in a patient with asthma
- Several comorbidities including pneumonia, diabetes and arrhythmias

Emergency Department Management of Asthma Exacerbations

- In the ED, the primary goals of asthma management include
 - correction of hypoxemia,
 - rapid improvement of airflow obstruction, and
 - prevention of progression or recurrence of symptoms.
- Interventions are based on
 - Clinical severity on arrival,
 - Response to initial therapy, and
 - Presence of risk factors of asthma morbidity and mortality

Box 9-6. Management of asthma exacerbations in acute care facility (e.g., emergency department)





- **Oxygen supplement**

- Oxygen should be administered and continued for at least 20 min after SABA administration to compensate for possible V/P abnormalities

- **Inhaled SABA**

- Main stay of emergent treatment
- Can be administered by intermittent nebulization, continuous nebulization or metered dose inhaler.
- Dosing
 - 0.5mg/kg/hr or
 - Based on body weight
 - 5-10kg 7.5mg/hr or 4 puffs
 - 10-20kg- 11.25mg/hr or 6 puffs
 - >20kg- 15mg/hr or 8 puffs

Mild asthma exacerbation

- SABA 4puffs to be given every 20 minutes for the first hr.
- If response lasts for 4-6 hrs; 4 puffs every 6 hrs for 24-48 hrs then PRN.
- If there is incomplete improvement; 4 puffs every 1hrly for 4–6 hrs,
- If good response; 4 puff every 4 to 6 hrly for 24–48 hrs then PRN.
- if necessary, oral or IV systemic corticosteroids.

Moderate asthma exacerbations

- SABA 4 puffs to be given every 20 min for the first hr.
- If response that lasts for > 3hrs, 4 puff every 4 hrly for 24 hrs, then every 6 hrly for 24–48 hrs then PRN.
- If there is incomplete improvement;
 - 4 puffs every 1 hrly for 4–6 hours, If there is good response 4 puff every 4 hourly for 24 hrs then every 6 hrly for 24 – 48 hrs then PRN. Or
 - **Ipratropium bromide** can be added; 0.25 to 0.50 mg doses, can be used every 20 min for 1hr, followed by the same dose range every 6 hrs for 48 hrs.
 - Fewer hospitalizations for adults but For children no benefits

- Administration of **systemic glucocorticoids** soon after arrival in the ED or after the first inhalation therapy is initiated
- Oral prednisolone 1-2mg/kg daily.
- In the ED, single oral, IV, or IM dose **dexamethasone** (0.3-0.6 mg/kg , max. 12 mg) has been found to be an effective alternative to **prednisone** and with a lower incidence of emesis.
- In addition, a second dose of dexamethasone should be given the next day whether discharged or admitted.
- Oral administration is as effective as IV

Severe Asthma Exacerbations

- **Indicators of a severe exacerbation include:**
 - Breathlessness,
 - Dyspnea,
 - Retractions, accessory muscle use,
 - Tachypnea or labored breathing,
 - Cyanosis,
 - Mental status changes,
 - Silent chest with poor air exchange,
 - Severe airflow limitation (PEF or FEV1 value <50% personal best or predicted values).

- ❖ Initial treatment includes supplemental oxygen, inhaled β -agonist therapy every 20 min for 1 hr, and oral or IV systemic corticosteroids

- **Oxygen Administration**

- **Nebulised albuterol/salbutamol:**

- 2.5 mg for children weigh less than 30 kg
- 5 mg for children weigh more than 30 kg, may be used diluted with 3 mL saline every 20 minutes for 1 hour
- Continuous nebulized albuterol is typically given at a dose of 0.5 mg/kg per hour.

- **Racemic/Nebulized epinephrine**

- 0.01ml/kg (maximum 0.5ml) diluted with 3ml NS every 20 minits for one hour

- **Systemic glucocorticoids**

- **Prednisolone** 1-2mg/kg daily for 3 - 5days. Or if the patient is unable to take po, IV
- **Hydrocortisone** 4mg/kg QID or administration of IV methylprednisolone 1mg/kg, in 2 divided doses can be given then change to PO prednisolone 1mg/kg/day when patient is able to tolerate orally to complete for 5 days

- **Nebulised ipratropium**

- 0.25 to 0.50 mg every 20 minutes for 1 hour, followed by the same dose range every 6 hours.
- The preferred method is to use Ipratropium bromide nebulized mixed with SABA

❖ Poor response to initial treatment, we recommend several therapies, including

- **Magnesium sulfate.** an initial bolus dose of 50 mg/kg (maximum dose, 2 g) infused for 20-60 min followed by continuous infusion dependent on the patient's weight.
 - weighing less than 30 kg may receive an infusion of 25 mg/kg/h and
 - weighing more than 30 kg may receive 20 mg/kg/h, although infusion rates must not exceed 2 g/h in any patient.
 - Can repeat once after 4-6 hrs

- **For patients who do not respond to these interventions**
 - Parenterally administered epinephrine: **Adrenaline** 0.01 mg/kg 1:1000 solution maximum dose of 0.5mg(0.5ml) administered SC or IM
 - Alternatively **terbutaline**(0.01 mL/kg of a 1 mg/mL solution) maximum dose of 0.4 mg (0.4 mL) intramuscularly or subcutaneously
 - Salbutamol IV: Bolus: 5microgram/kg/min for first hour then consider Infusion: 1-2 microgram/kg/min
 - **Iv aminophylline**: combination of theophylline and ethylenediaminen
 - **Inhaled heliox** (helium and oxygen mixture)

Criteria for Discharge

- No longer requires **supplemental O₂**
- On a **treatment regimen feasible at home**
- Observed at least **one interval** on home-equivalent regimen
- **Proper MDI-spacer technique** demonstrated
- **FEV1 >70% predicted** (if spirometry available)
- **Access to meds + follow-up** confirmed
- **Asthma education completed**

Discharge Medications

- **SABA (albuterol):** 2 inhalations as needed for symptom relief, not regularly
- **Systemic steroids:**
 - Prednisone/Prednisolone × 3–5 days
 - Or Dexamethasone × 2 days
 - May extend up to 10 days in high-risk cases
 - ≤10 days → **no taper needed**
- a moderate or severe exacerbation, ICS should be started

Asthma Action Plan

- Daily meds + timing
- Rescue meds + symptom
- Emergency contact phone number
- List of triggers and avoidance

Follow-up

- follow-up appointment within 1–3 days and another within 2–3 month depending on the clinical, social and practical context of the exacerbation

Prognosis

- Recurrent coughing and wheezing occurs in 35% of preschool-age children.
- Of these, approximately one third continue to have persistent asthma into later childhood, and approximately two thirds improve on their own through their teen years.
- Asthma severity by ages 7-10 yr is predictive of asthma persistence in adulthood.
- Children with moderate to severe asthma and with lower lung function measures are likely to have persistent asthma as adults.
- Children with milder asthma and normal lung function are likely to improve over time.

Prevention

- **Prevention**
- Early immunomodulatory intervention might prevent asthma development.
- A hygiene hypothesis supports that naturally occurring microbial exposures in early life might drive early immune development.
- Several non pharmacologic therapeutic measures with numerous positive health attributes.
- Avoidance of environmental tobacco smoke (beginning prenatally), prolonged breastfeeding (>4 months), an active lifestyle.
- Immunizations are currently not considered to increase the likelihood of development of asthma.

References:

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THANK YOU

09/13/2026